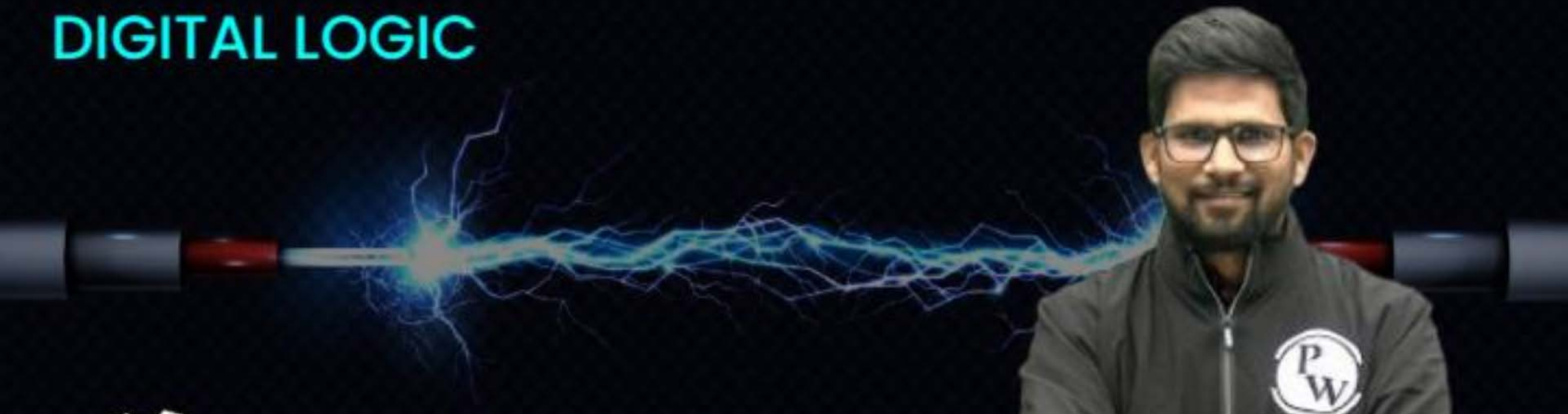
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 10

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



- Synch. Counter
-
-



- Shift Register
- Question Discussion

Shift Register

For what purpose shift registers are used ?

→ to store data or information.

- Types of shift registers :

→ SISO

→ SIPO

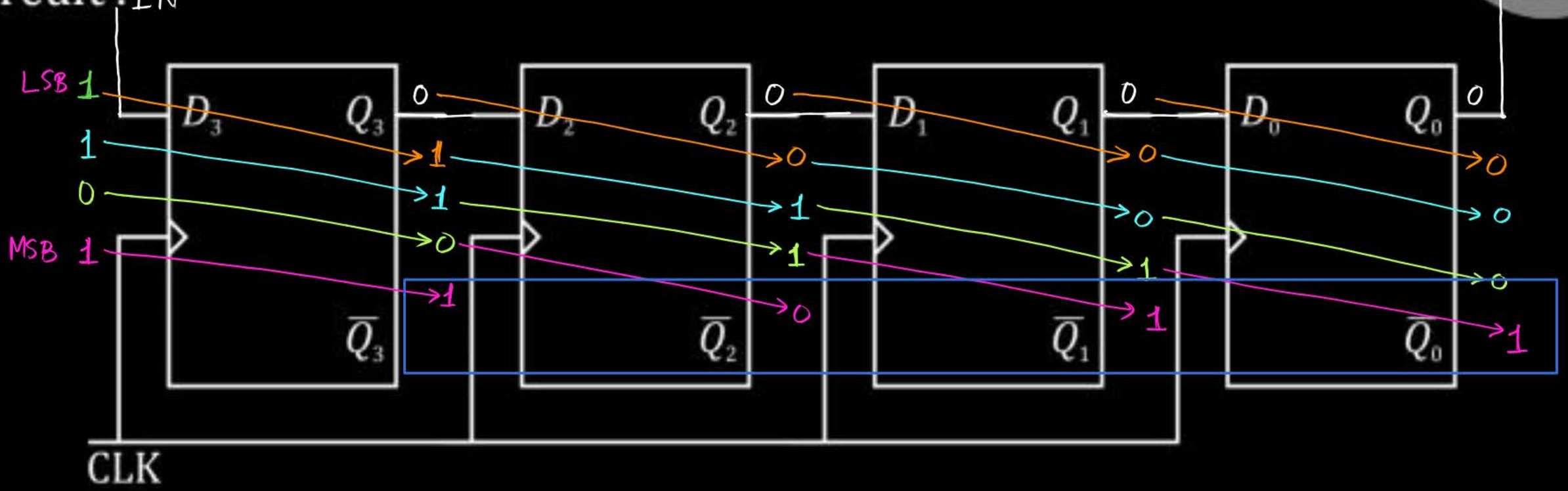
→ PIPO

→ PISO

1. SISO : 4 bit

1011

Circuit :



1st

2nd

3rd

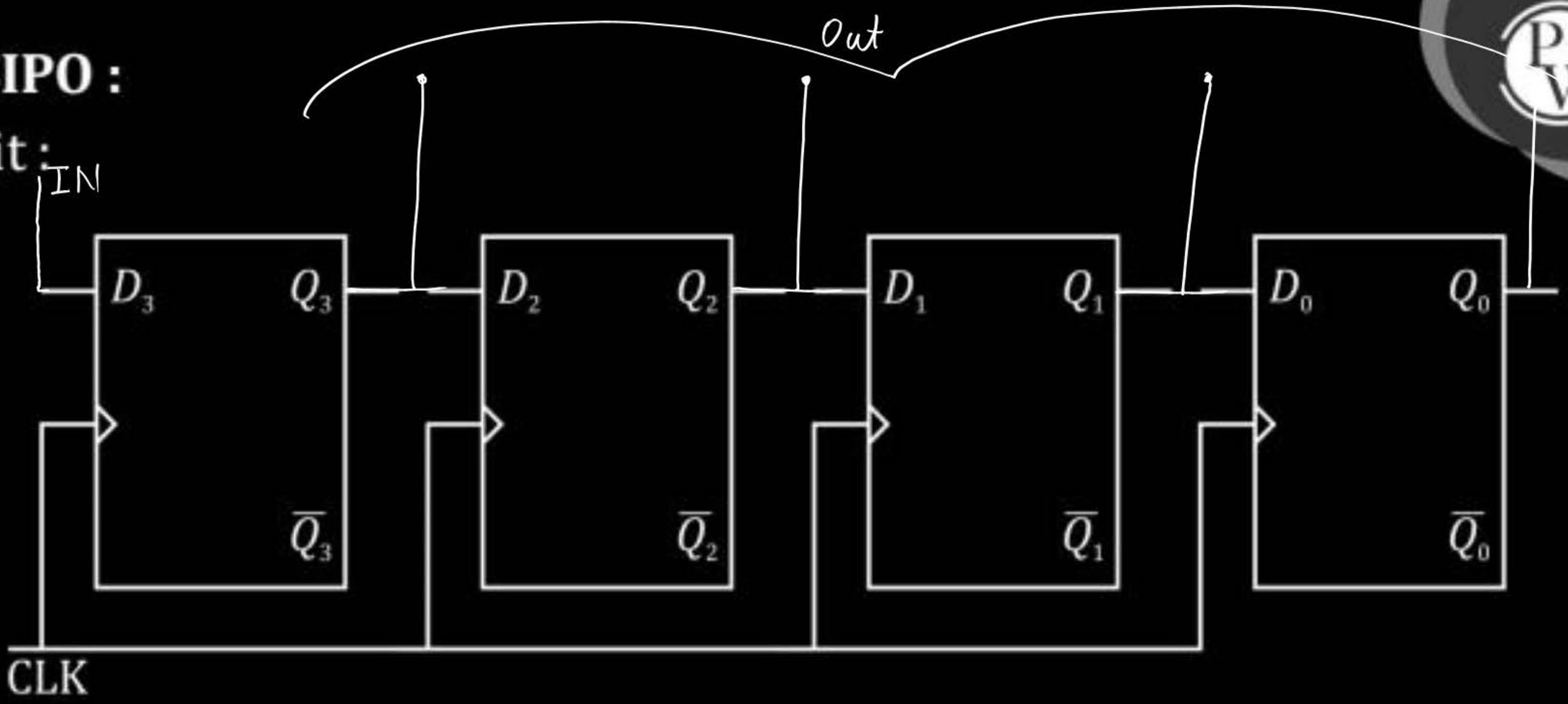
4th

IMP Points :

- To store n-bit data serially, we require n-clock pulses [n-Tclk].
- To read n-bit data serially, we require (n-1) clock pulses [(n-1) Tclk].

2. SIPO :

Circuit :



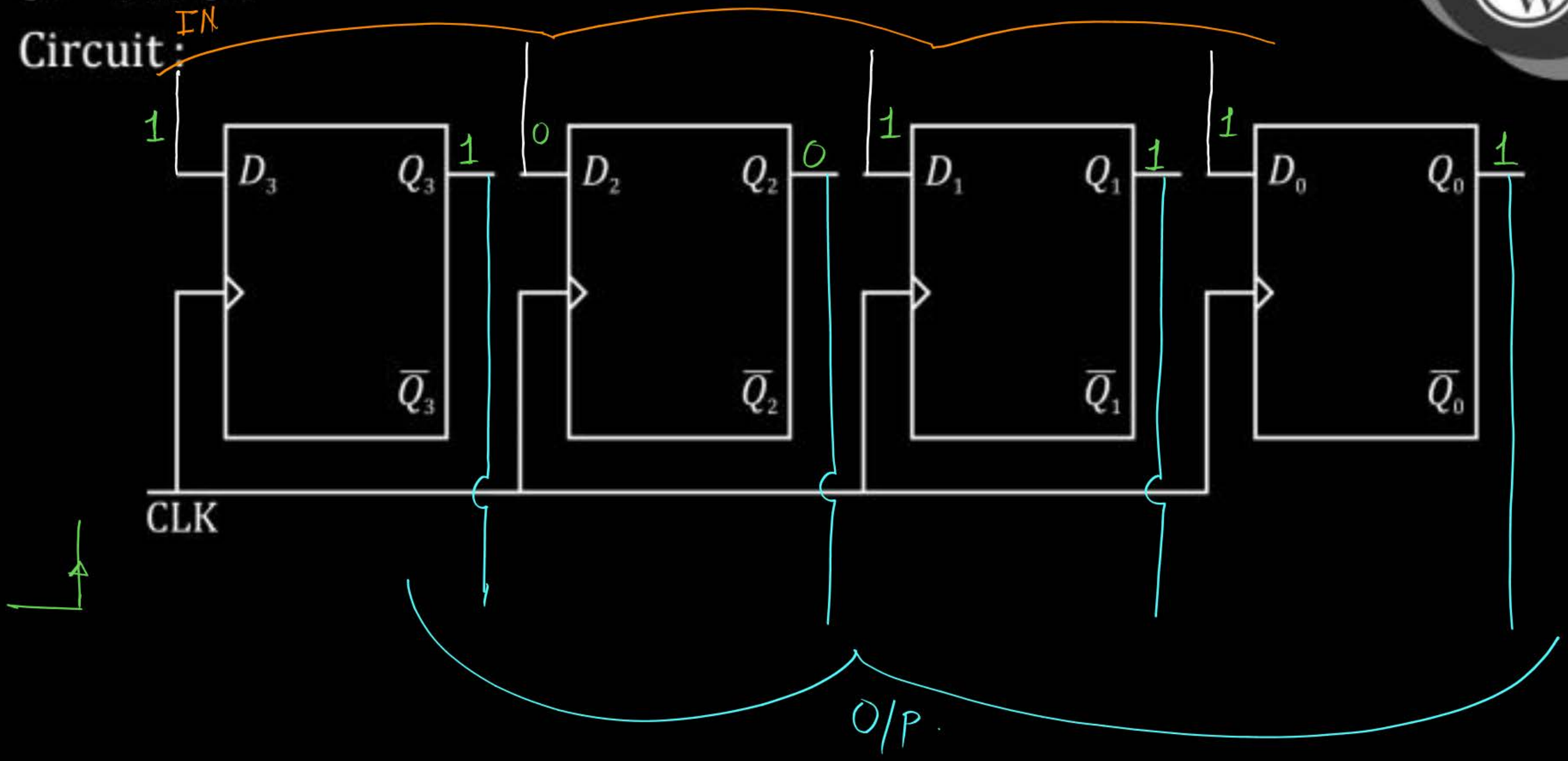
IMP Points :

- To serially-IN , n-bit data , we require n-clock pulses [n-Tck].
- To parallel-out, n-bit data , we require 0-clock pulses [0-Tck].

3. PIPO :

1011

Circuit :

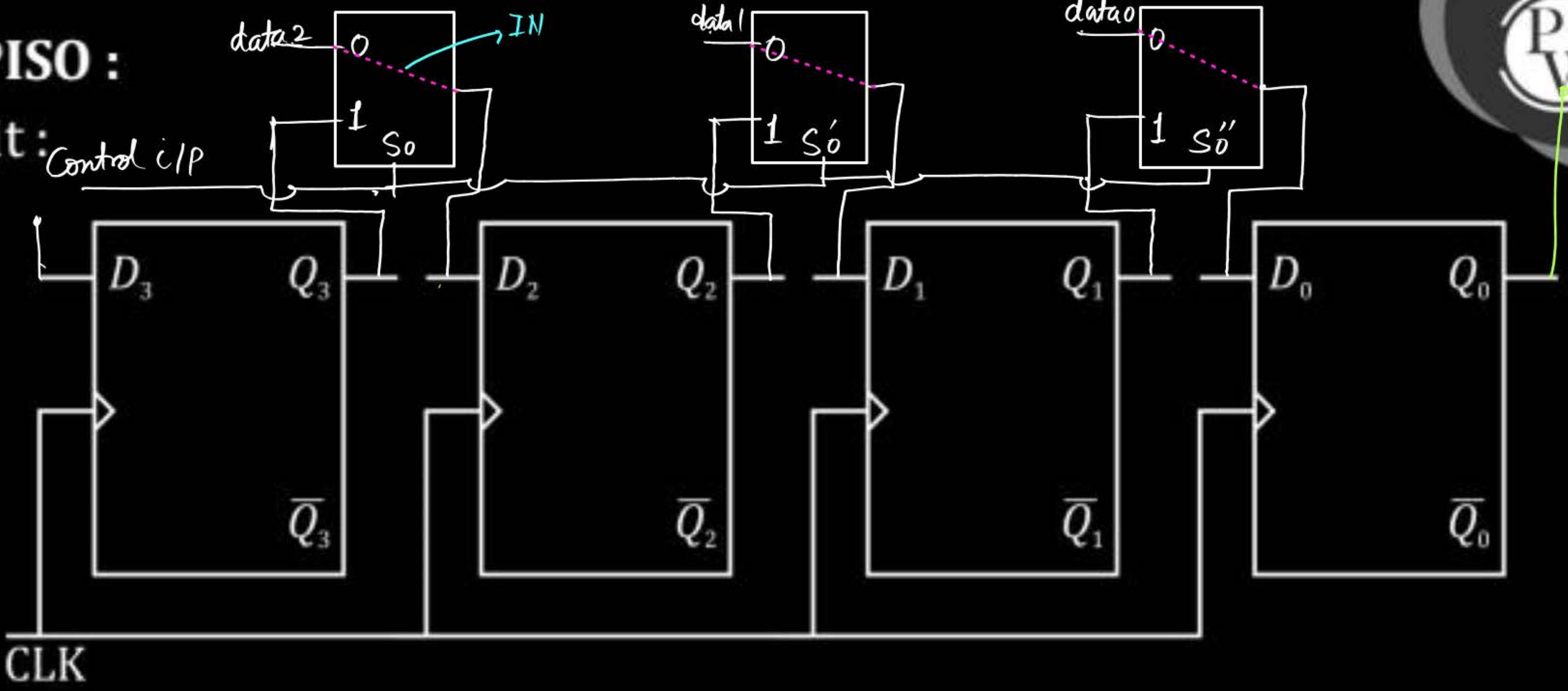


IMP Points :

- To store n-bit data parallely we require 1-clock pulses [1-Tclk].
- To read n-bit data parallely, we require 0-clock pulses [0-Tclk].

4. PISO :

Circuit :



Control i/P = 0 ⟶ for storing the data

Control i/P = 1 ⟶ for read operation.

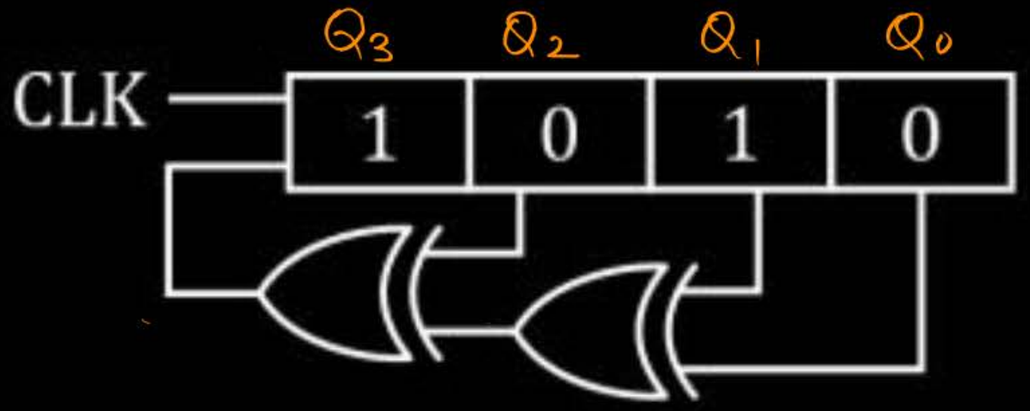
IMP Points :

- To store n-bit data parallely we require 1-clock pulse [1-Tclk].
- To read n-bit data serially we require (n-1) clock pulses [(n-1) Tclk].

Register	IN	OUT
SISO	n-Tick	$(n-1)$ Tick
SIPO	n-Tick	0-Tick
PIPO	1-Tick	0-Tick
PISO	1-Tick	$(n-1)$ Tick.

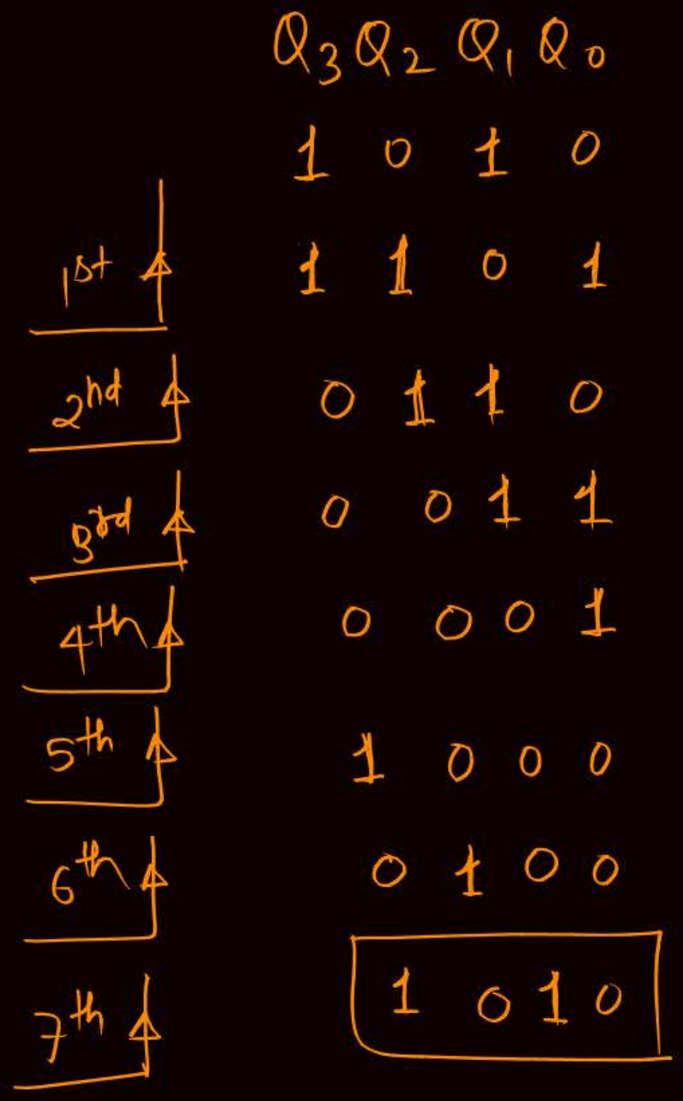
Question

A four bit SIPO register is shown in fig. with starting state $(1010)_2$.



$Q_3(n+1) = D_3 = Q_2(n) \oplus Q_1(n) \oplus Q_0(n)$

After application of how many clock pulses content of SIPO will again will be $(1010)_2$ 7


$Q_3 Q_2 Q_1 Q_0$
1 0 1 0
1st
1 1 0 1
2nd
0 1 1 0
3rd
0 0 1 1
4th
0 0 0 1
5th
1 0 0 0
6th
0 1 0 0
7th
1 0 1 0

Question

A 4-bit shift register circuit configured the right shift operation i.e. D_{in} – A, A – B, B – C, C – D is shown. If the present state of shift register is ABCD = $(1101)_2$, then the no. of clock cycle required to reach the state ABCD = $(1111)_2$ _____.



$Q_A(n+1) = D(in) = Q_A(n) \oplus Q_D(n)$

	Q_A	Q_B	Q_C	Q_D
	1	1	0	1
1st	0	1	1	0
2nd	0	0	1	1
3rd	1	0	0	1
4th	0	1	0	0
5th	0	0	1	0
6th	0	0	0	1
7th	1	0	0	0
8th	1	1	0	0
9th	1	1	1	0
10th	1	1	1	1

Question

Design a counter that follows the sequence :

0 – 1 – 3 – 2 – 1 – 0 → using D-FF

starting state

ff ⟶ 3 ff.

0	0	0
0	0	1
0	1	1
0	1	0
1	0	1

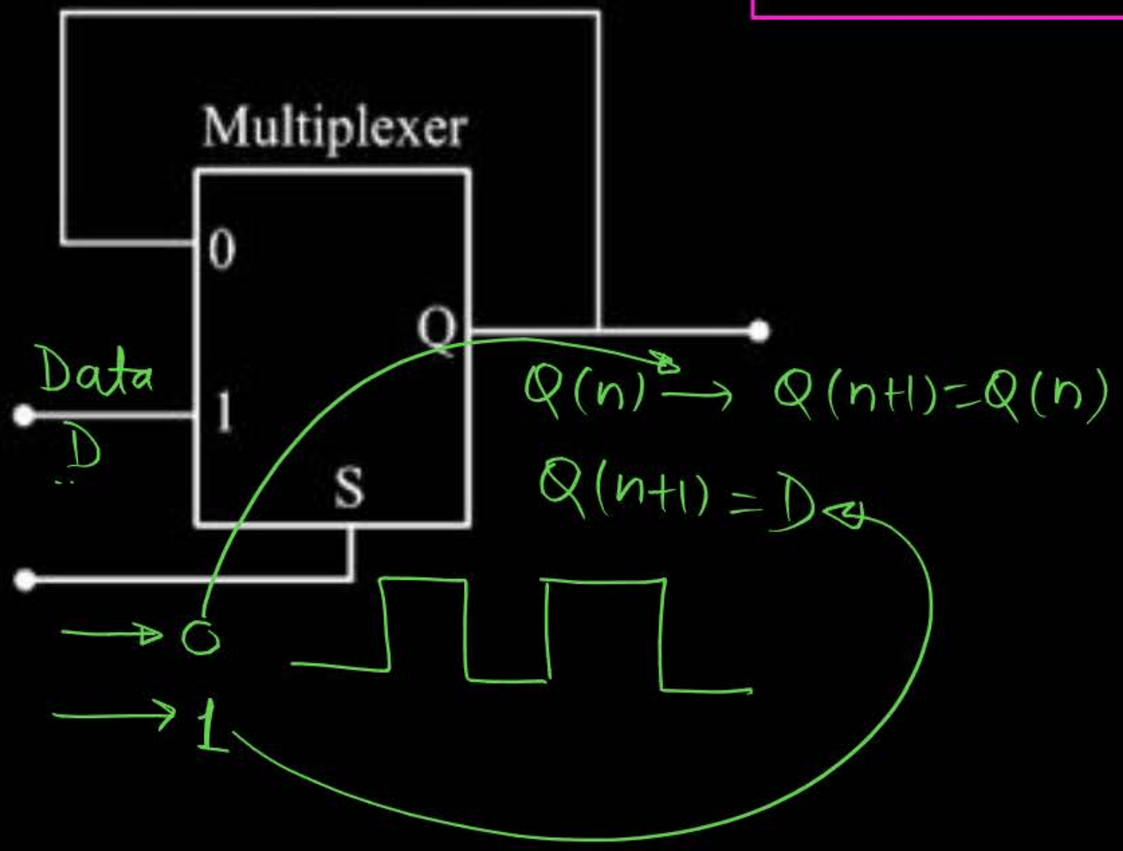
[MCQ]

The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.

Match the function equivalence of this circuit to one of the following options.

[GATE-2023-CS: 1M]

+ve level triggered D-FF



A. D Flip-flop

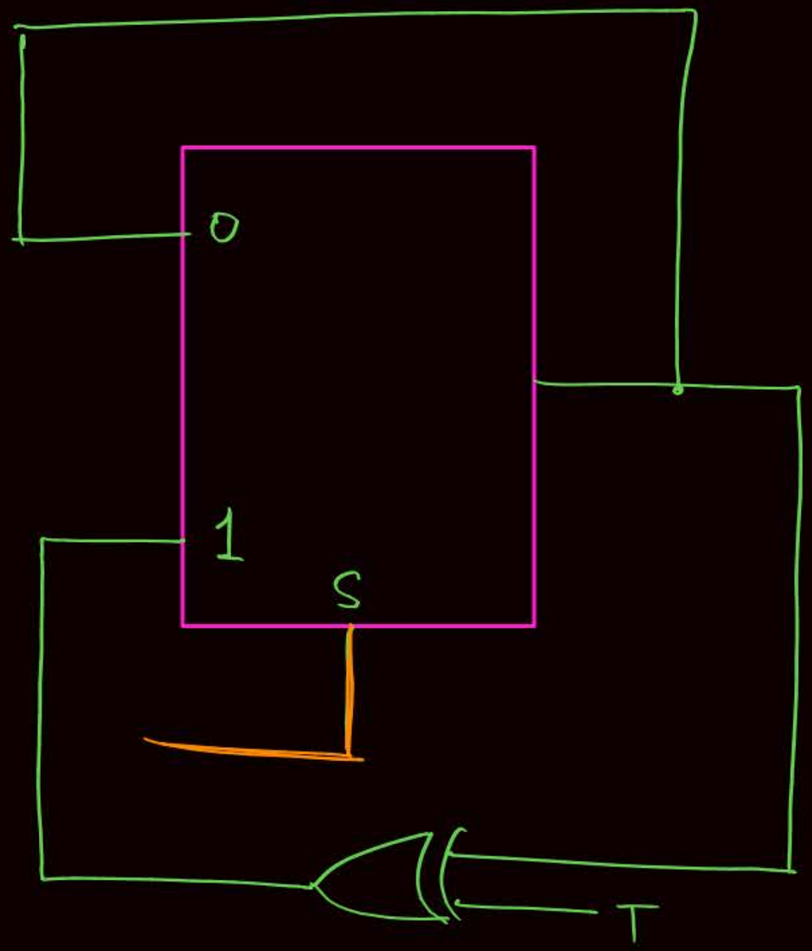
B. D Latch

C. Half-adder

D. Demultiplexer

$S=1 \longrightarrow Q(n+1) = T \oplus Q(n)$

$S=0 \longrightarrow Q(n+1) = Q(n)$



#Q. Given below is the diagram of a synchronous sequential circuit with one J-K flip-flop and one T flip-flop with their outputs denoted as A and B respectively, with $J_A = (A' + B')$, $K_A = (A + B)$, and $T_B = A$.
Starting from the initial state (AB = 00), the sequence of states (AB) visited by the circuit is

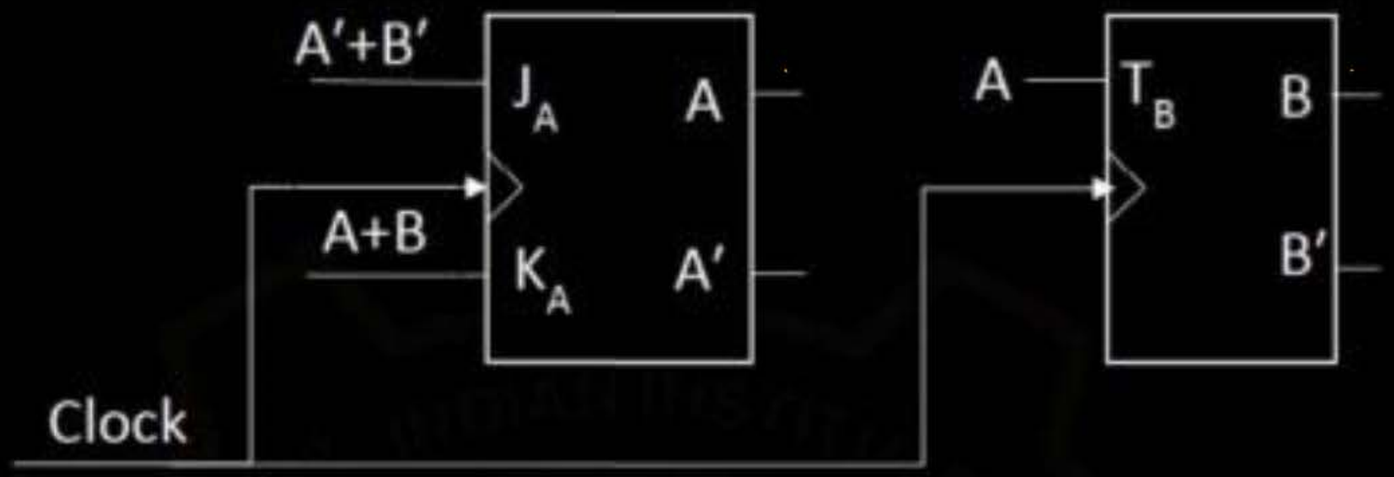
H.W.

A 00 – 01 – 10 – 11 – 00

B 00 – 10 – 01 – 11 – 00

C 00 – 10 – 11 – 01 – 00

D 00 – 01 – 11 – 00



[MCQ]

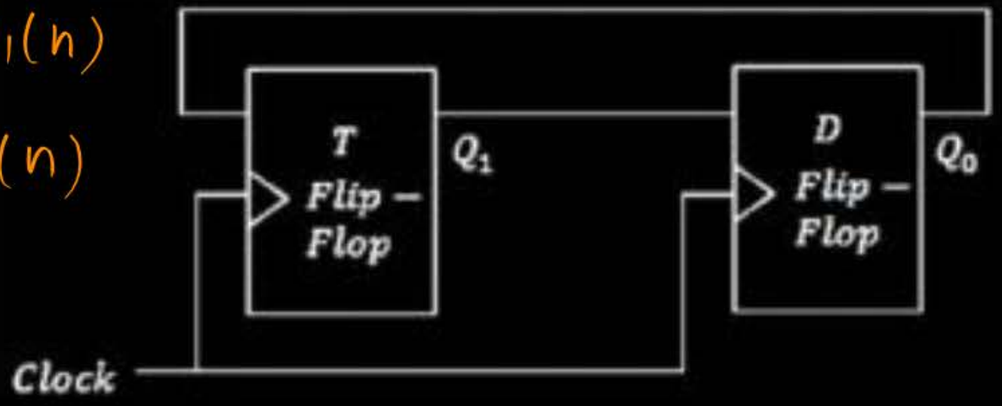
Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.

Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs

$Q_1(n+1) = Q_0(n) \oplus Q_1(n)$

$Q_0(n+1) = D_0 = Q_1(n)$

[GATE-2017-CS: 1M]



$Q_1\ Q_0$

1 1

1st ↑ → 0 1

2nd ↑ → 1 0

3rd ↑ → 1 1

4th ↑ → 0 1

(A) Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 00 respectively

(B) Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 01 respectively

(C) Q_1Q_0 after the 3rd cycle are 00 and after the 4th cycle are 11 respectively

(D) Q_1Q_0 after the 3rd cycle are 01 and after the 4th cycle are 01 respectively.

[MCQ]

A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flip-flop as follows. The Q output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that J = K = 1 is the toggle mode and J = K = 0 is the state-holding mode of the JK flip-flop. Both the flip-flops have non-zero propagation delays. [GATE-2015-CS: 1M]

(A) 0110110...

(B) 0100100...

(C) 011101110...

(D) 011001100...

$Q_1 Q_0$

0 1

1 0

1 1

0 1

$Q_1(n+1) = Q_0(n)\,\overline{Q_1}(n) + \overline{Q_0}(n)\cdot Q_1(n) = Q_0(n) \oplus Q_1(n)$

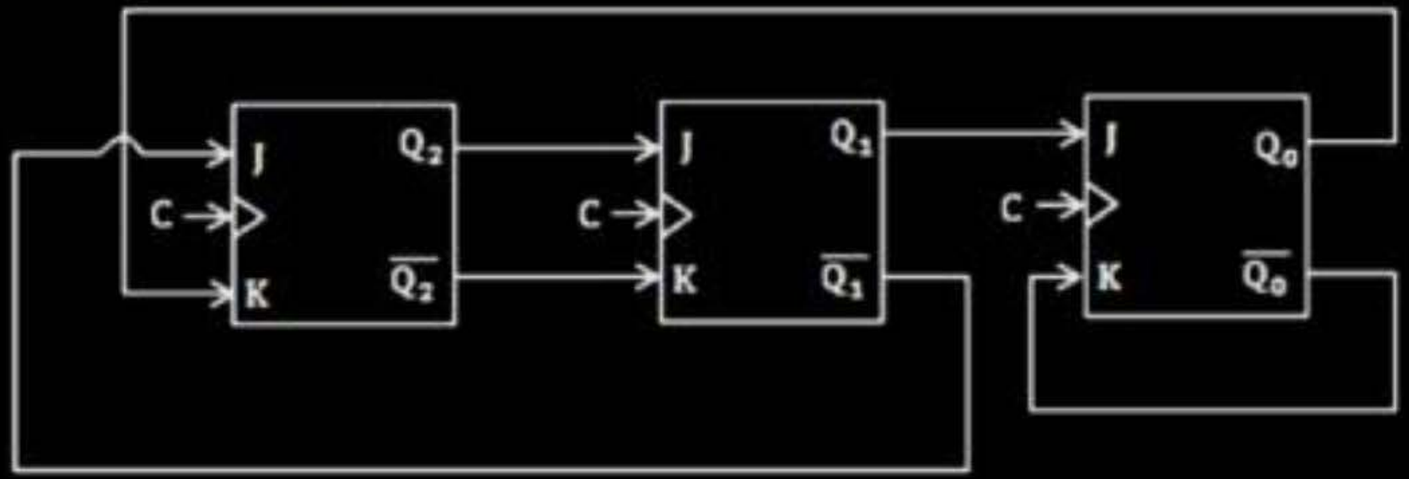
$Q_0(n+1) = D_0 = Q_1(n)$

0 1 1 0 1 1 0 1 1

[MCQ]

The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is :

[GATE-2014-CS: 1M]



H.W.

A. 001, 010, 011

B. 111, 110, 101

C. 100, 110, 111

D. 100, 011, 001

[MCQ]

The minimum number of D flip-flops needed to design a mod-258 counter is :

[GATE-2011-CS: 1M]

H.W

A 9

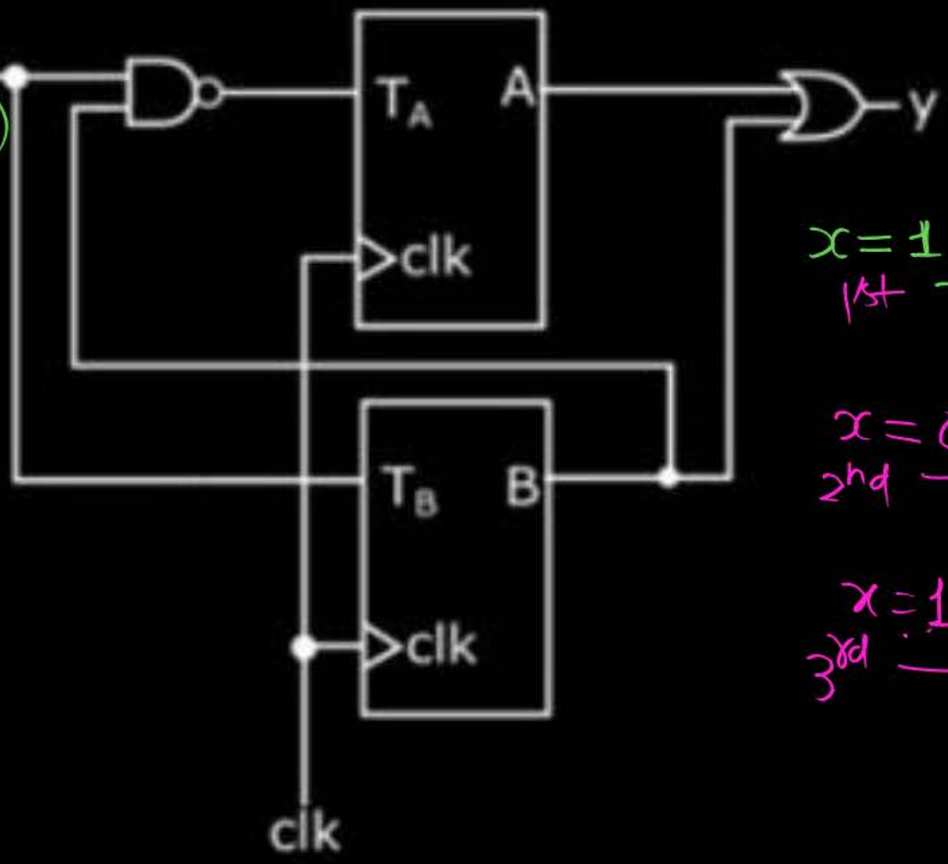
B 8

C 512

D 258

#Q. Two T-flip flops are interconnected as shown in the figure. The present state of the flip flops are: A = 1, B = 1. The input x is given as 1, 0, 1 in the next three clock cycles. The decimal equivalent of $(ABy)_2$ with A being the MSB and y being the LSB, after the 3rd clock cycle is $(7)_{10}$

$$Q_A(n+1) = T_A \oplus Q_A(n)$$
$$= \overline{x \cdot Q_B(n)} \oplus Q_A(n)$$
$$Q_B(n+1) = T_B \oplus Q_B(n)$$
$$= x \oplus Q_B(n)$$



A	B	y
1	1	1
1	0	1
0	0	0
1	1	1

x = 1 (1st)

x = 0 (2nd)

x = 1 (3rd)

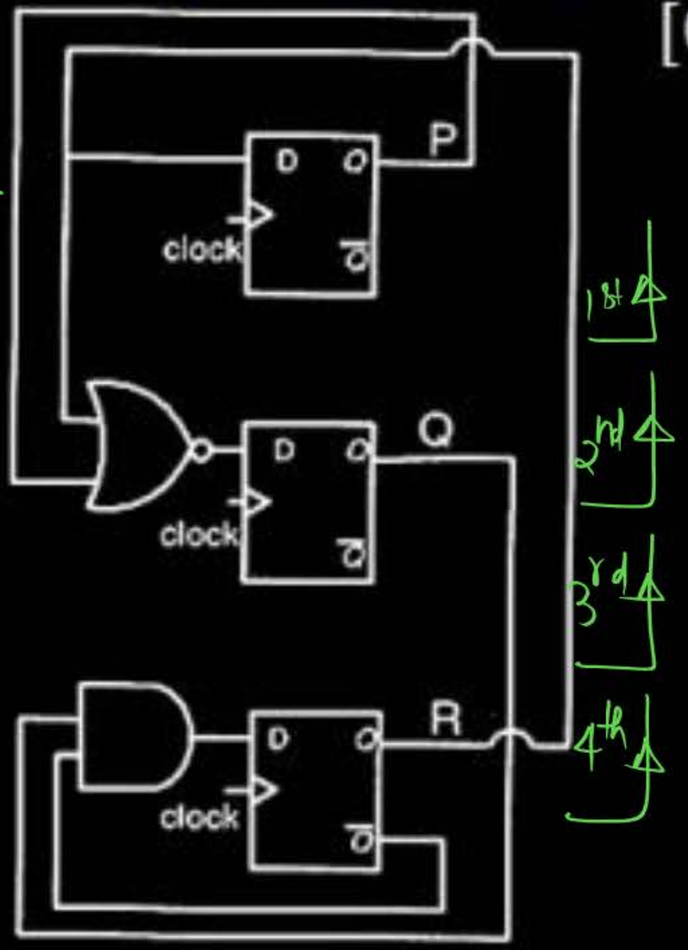
[MCQ]

Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration. If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter ? [GATE-2011-CS: 1M]

$Q_P(n+1) = D_P = Q_R(n)$

$Q_Q(n+1) = D_Q = \overline{Q_P(n) + Q_R(n)}$

$Q_R(n+1) = D_R = Q_Q(n) \cdot \overline{Q_R}(n)$



P	Q	R
0	0	0
0	1	0
0	1	1
1	0	0
0	0	0

MOD 4

A. 3

B. 4

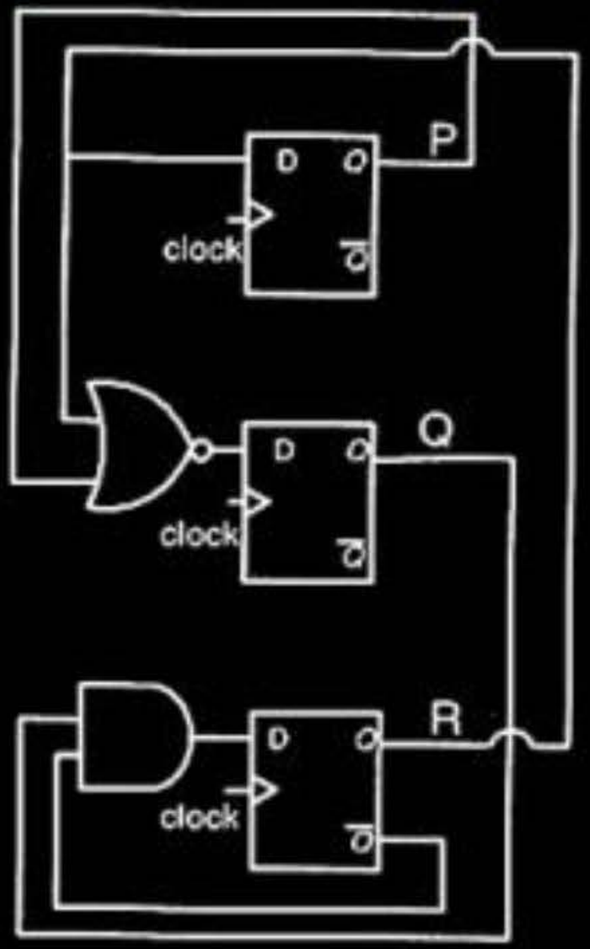
C. 5

D. 6

[MCQ]

Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration. If at some instance prior to the occurrence of the clock edge, P, Q and R have a value 0, 1 and 0 respectively, what shall be the value of PQR after the clock edge? [GATE-2011-CS: 1M]

H.W.



(A) 000

(B) 001

(C) 010

(D) 011

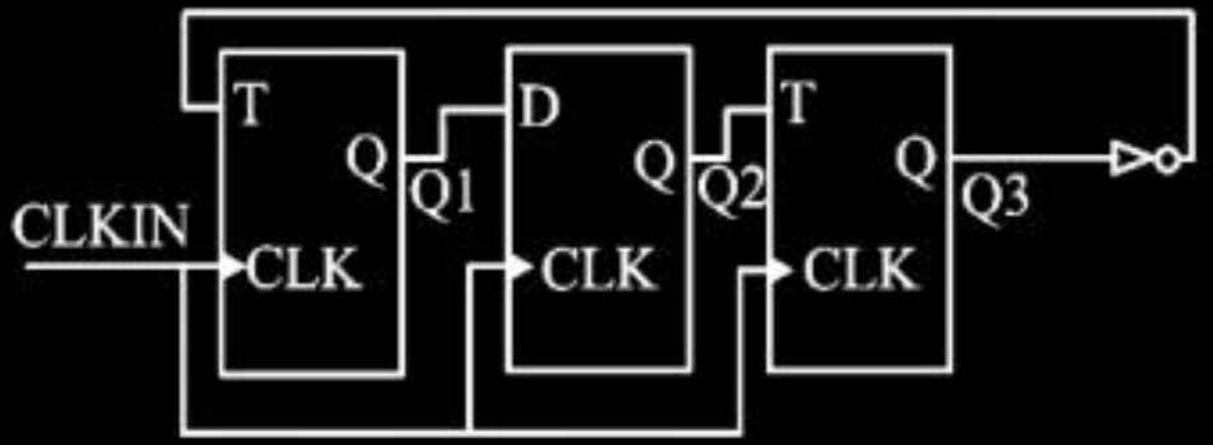
[MCQ]

Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1 and 1, respectively.

Which one of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?

[GATE-2023-CS: 2M]

H.W.



A. (0, 0, 1)

B. (1, 0, 0)

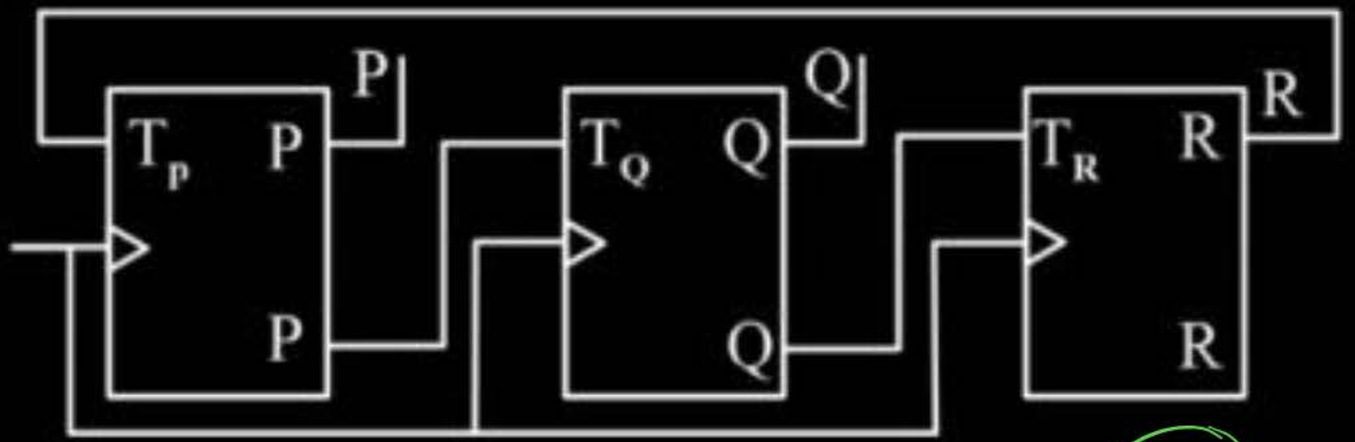
C. (1, 0, 1)

D. (1, 1, 1)

[MCQ]

Consider a 3-bit counter, designed using T flip-flops, as shown below. Assuming the initial state of the counter given by PQR as 000. What are the next three states ?

[GATE-2021-CS: 1M]



HW

(A) 011,101,000

(B) 001,010,111

(C) 011,101,111

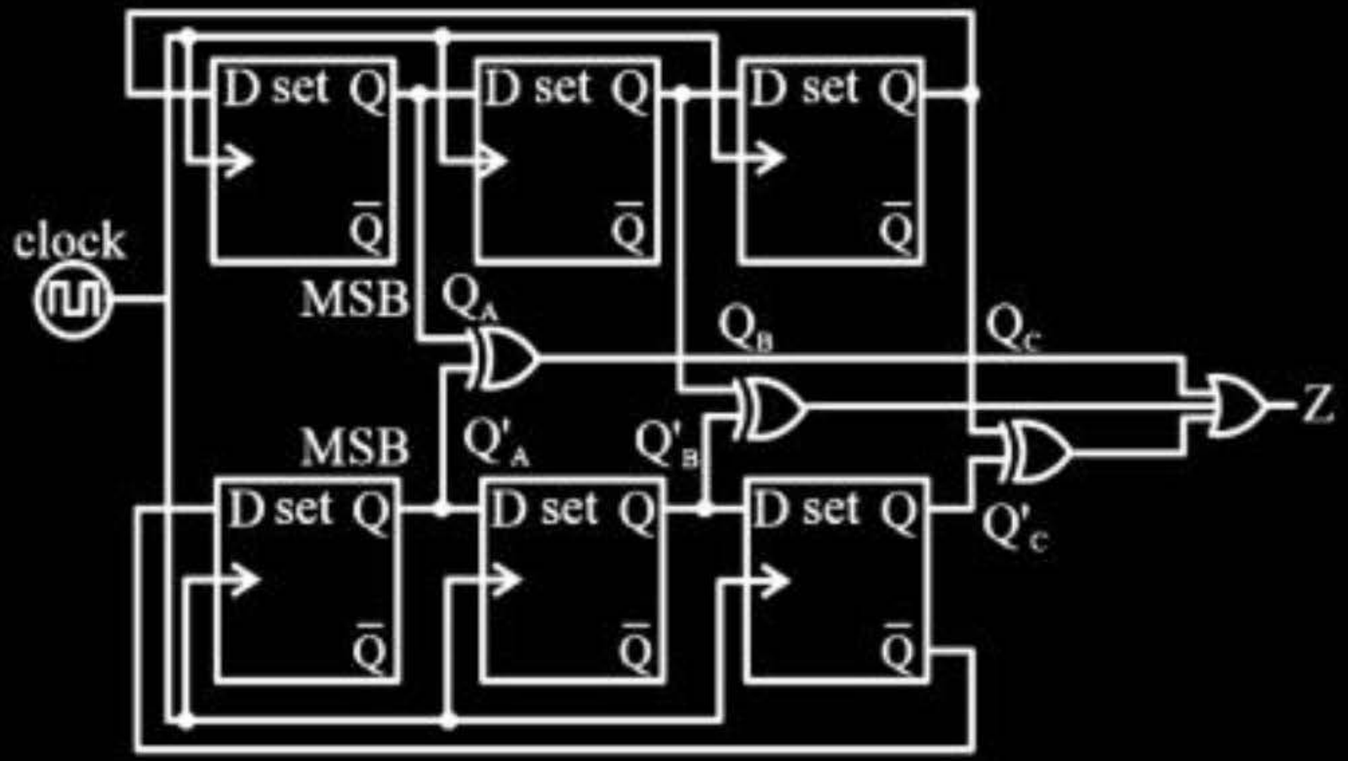
(D) 001,010,000

#Q. For the synchronous sequential circuit shown below, the output Z is zero for the initial conditions

$$Q_A Q_B Q_C = Q'_A Q'_B Q'_C = 100$$

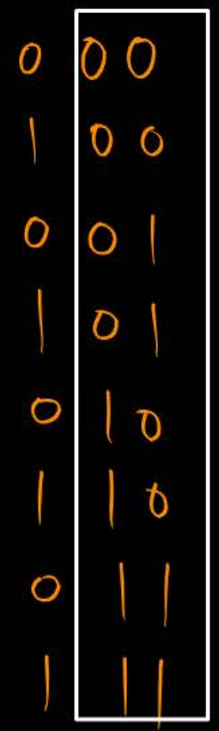
The minimum number of clock cycles after which the output z would again become zero is _____.

H.W.



[NAT]

The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0,0,1,1,2,2,3,3,0,0....)is. [GATE-2021-CS: 2M]



→ design it [H.W.]

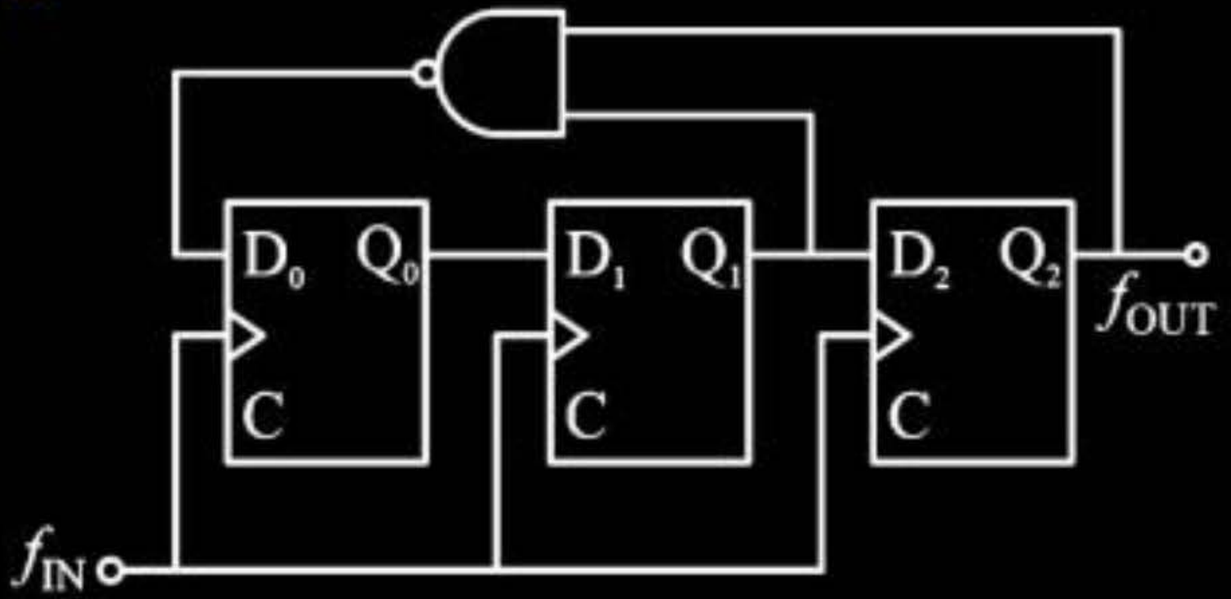
3 → 2 ffs

for 2-different comb → 1 bit → 1 ff

↓

3ff

#Q. Which one of the following statements is true about the digital circuit shown in the figure?



H.W.

A It can be used for dividing the input frequency by 3.

B It can be used for dividing input frequency by 5.

C It can be used for dividing the input frequency by 7.

D It cannot be reliably used as a frequency divider due to disjoint internal cycles.

[MCQ]

The next state table of a 2-bit saturating up-counter is given below.
The counter is built as a synchronous sequential circuit using T flip-flops.
The expressions for T_1 and T_0 are [GATE-2017-CS: 1M]

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

T_1 T_0

H.W.

(A) $T_1 = Q_1Q_0, T_0 = \overline{Q_1} + \overline{Q_0}$

(B) $T_1 = \overline{Q_1}Q_0, T_0 = \overline{Q_1} + \overline{Q_0}$

(C) $T_1 = Q_1 + Q_0, T_0 = \overline{Q_1} + \overline{Q_0}$

(D) $T_1 = \overline{Q_1}Q_0, T_0 = Q_1 + Q_0$

[MCQ]

Consider a 4-bit Johnson counter an initial value of 0000. The counting sequence of this counter is -

[GATE-2015-CS: 1M]

0000

1000

A 0,1,3,7,15,14,12,8,0

B 0,1,3,5,7,9,11,13,15,0

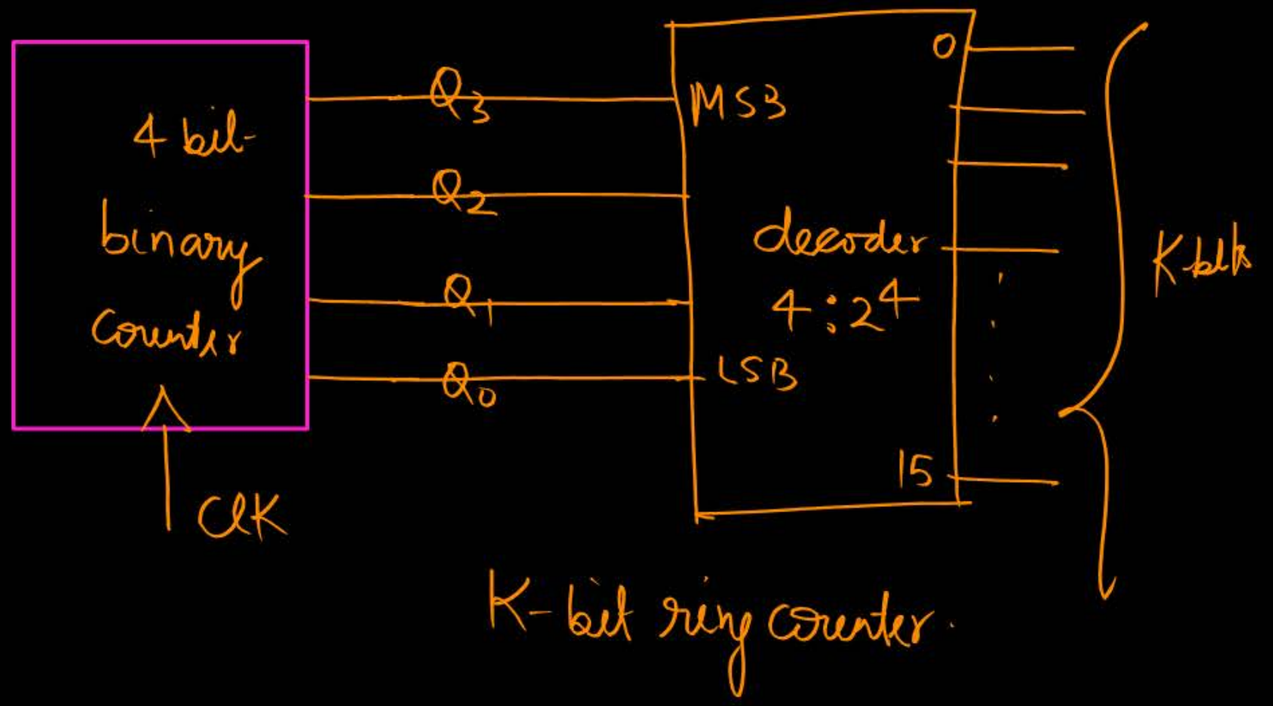
C 0,2,4,6,8,10,12,14,0

D 0,8,12,14,15,7,3,1,0

[MCQ]

Let $k = 2^n$. A circuit is built by giving the output of an n-bit binary counter as input to an n-to- 2^n bit decoder. This circuit is equivalent to a

[GATE-2014-CS: 1M]



(A) k-bit binary up counter.

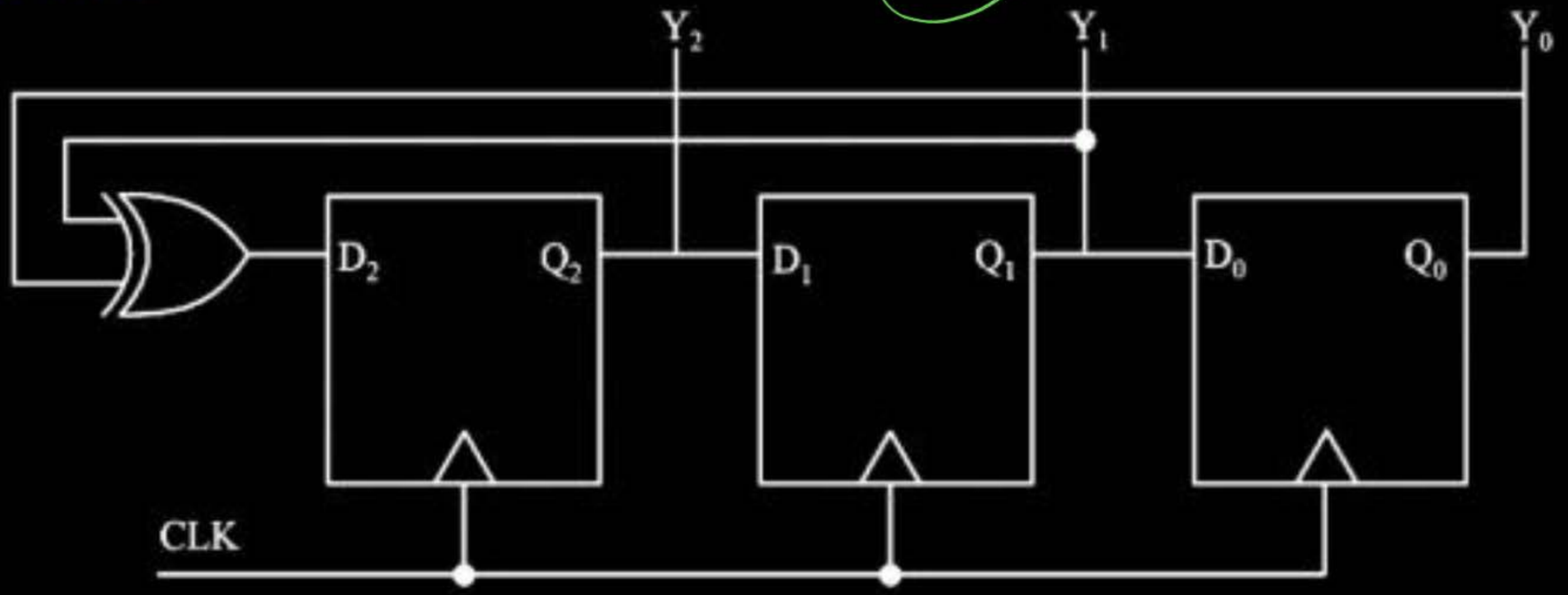
(B) k-bit binary down counter.

(C) k-bit ring counter.

(D) k-bit Johnson counter.

#Q. A three-bit pseudo random number generator is shown. Initially the value of output $Y = Y_2\ Y_1\ Y_0$ is set to 111. The value of output Y after three clock cycles is

H.W.



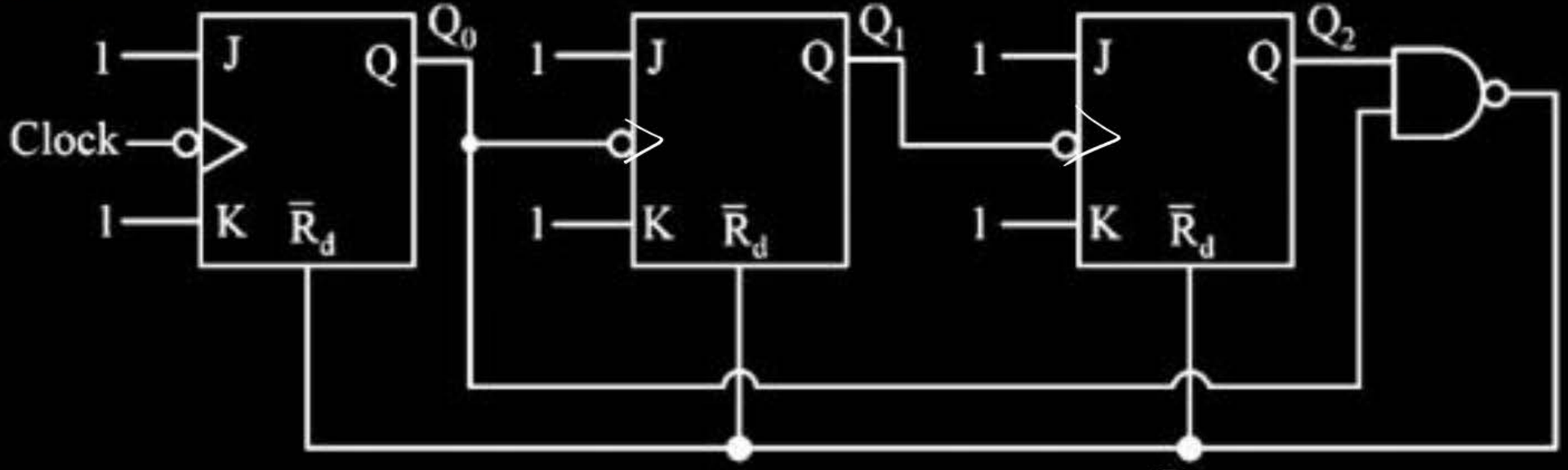
A 000

B 001

C 010

D 100

#Q. The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset ($\overline{R_d}$ input). The counter corresponding to this circuit is



H.W.

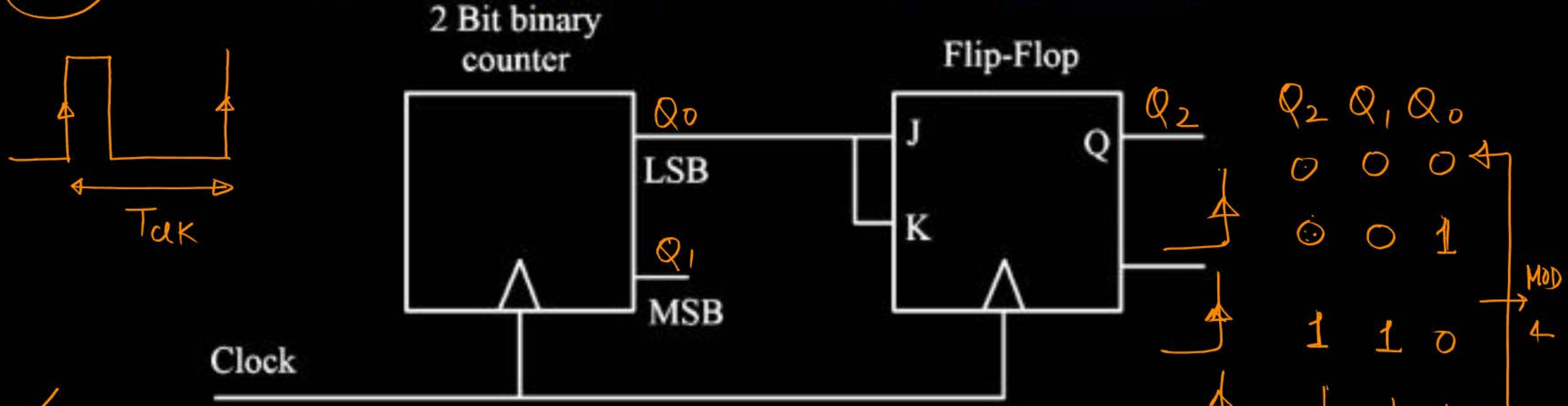
A a modulo-5 binary up counter

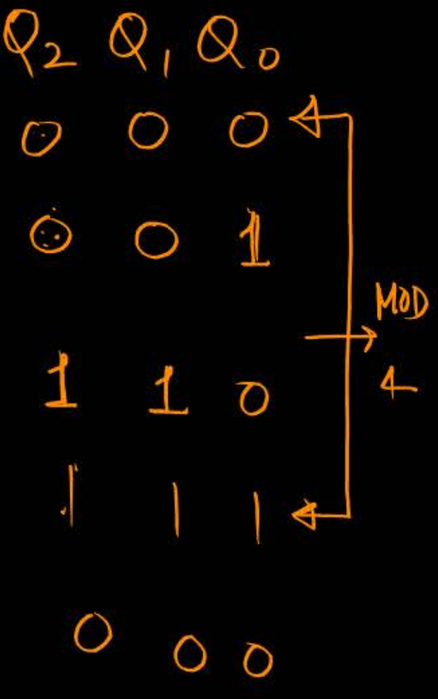
B a modulo-6 binary down counter

C a modulo-5 binary down counter

D a modulo-6 binary up counter

#Q. For the circuit shown, the clock frequency is f_0 and the duty cycle is 25%. For the signal at the Q output of the Flip-Flop, _____.





$Q_2(n+1) = Q_0(n) \oplus Q_2(n)$

$Q_2 \rightarrow T_{OFF} = 2T_{clk}$

$T_{ON} = 2T_{clk}$

$T_{Q_2} = 4T_{clk} \rightarrow f_{Q_2} = f_{clk}/4$

- [A] frequency is $f_0/4$ and duty cycle is 50%
- [B] frequency is $f_0/4$ and duty cycle is 25%
- [C] frequency is $f_0/2$ and duty cycle is 50%
- [D] frequency is f_0 and duty cycle is 25%

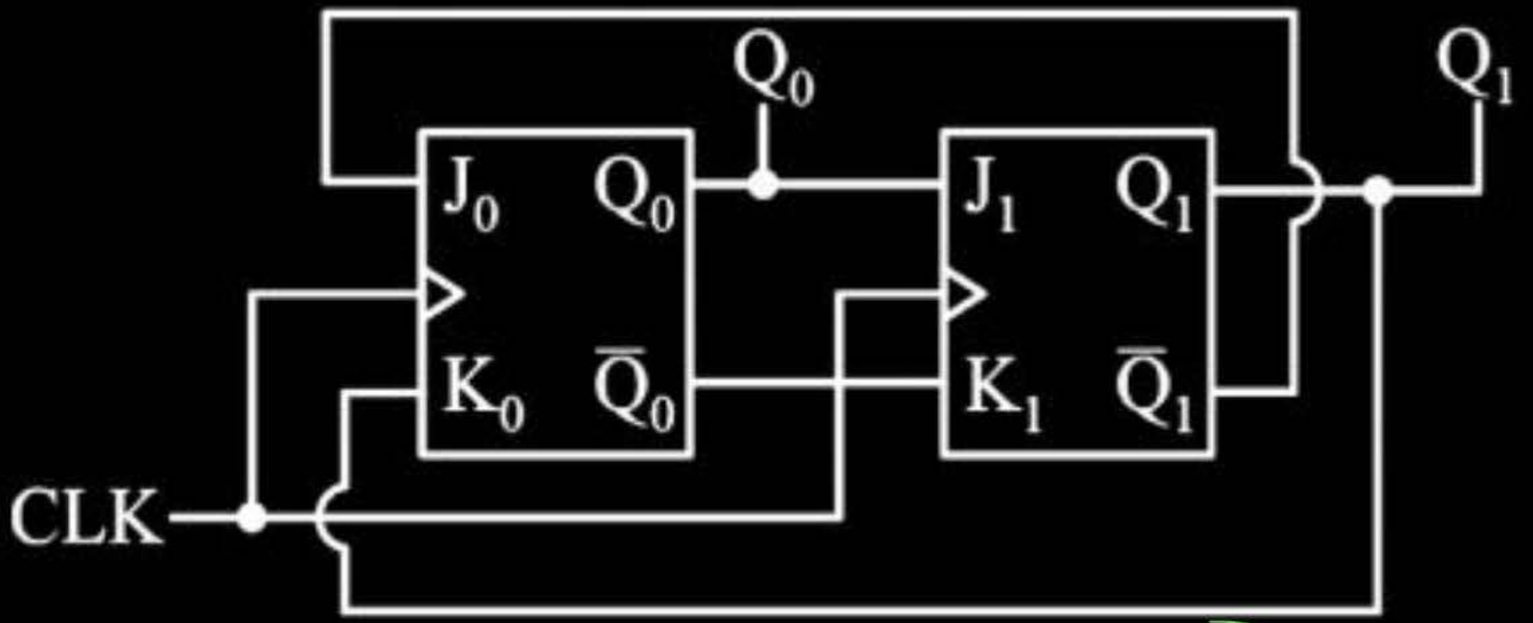
#Q. In the following sequential circuit, the initial state (before the first clock pulse) of the circuit is $Q_1\ Q_0 = 00$. The state $(Q_1\ Q_0)$, immediately after the 333rd clock pulse is

A 00

B 01

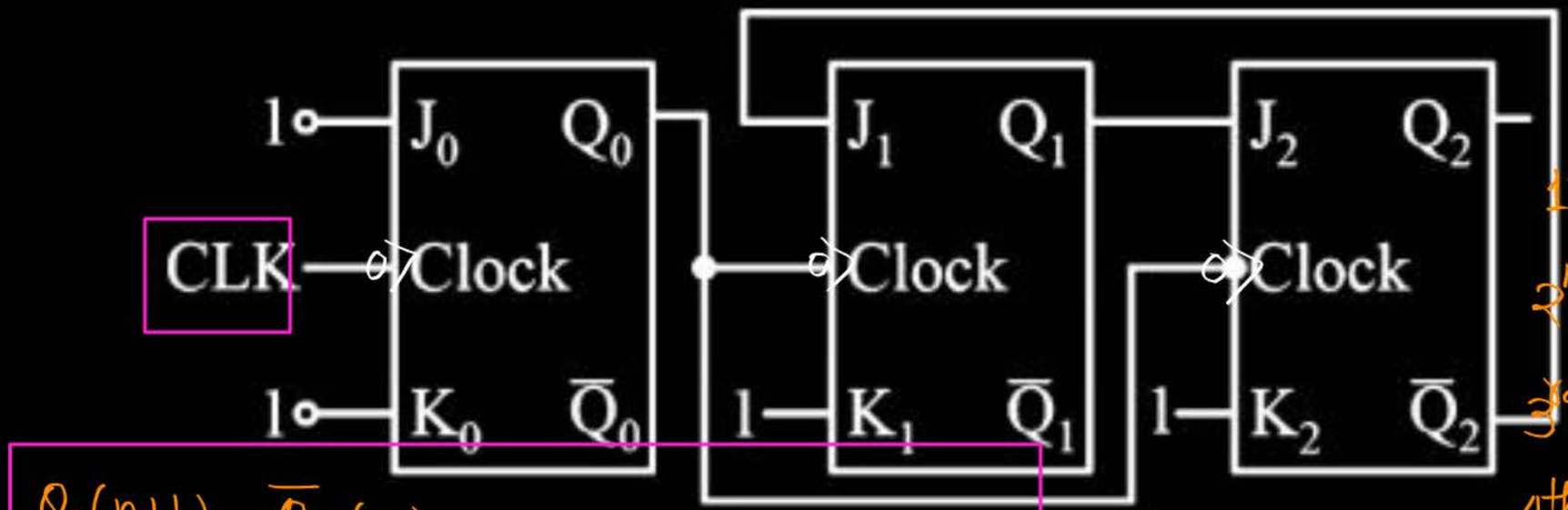
C 10

D 11



17·W

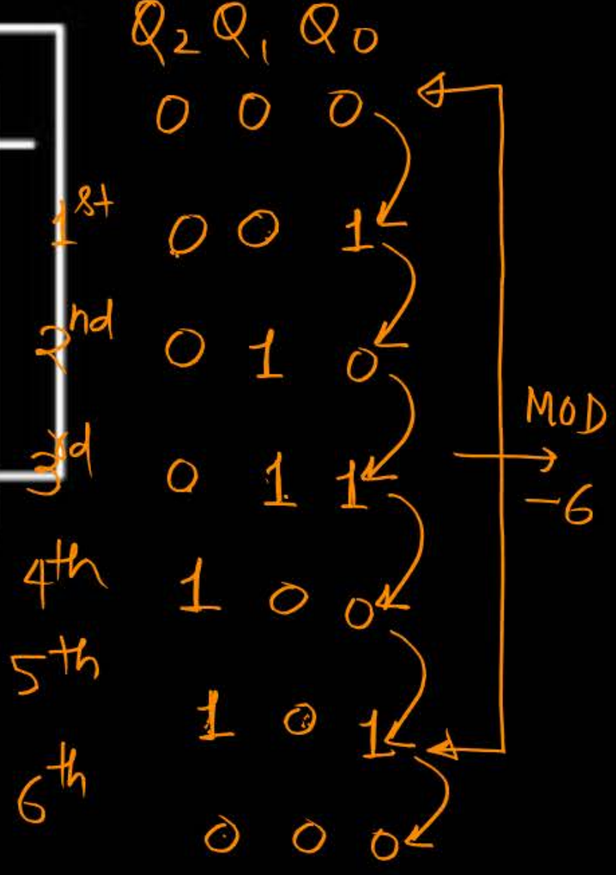
#Q. The figure shows a digital circuit constructed using negative edge triggered J-K flip flops. Assume a starting state of $Q_2\ Q_1\ Q_0 = 000$ will repeat after 6 number of cycles of the clock CLK



$Q_0(n+1) = \overline{Q}_0(n)$

$Q_1(n+1) = \overline{Q}_2(n) \cdot \overline{Q}_1(n) = \overline{Q_1(n) + Q_2(n)}$

$Q_2(n+1) = Q_1(n) \cdot \overline{Q}_2(n)$





Topic : 2 Min Summary

- Shift register
- Question Discussion

Thank you

GW

Soldiers !

